

Heliostat Wiring Schematic Summary

Power: Solar array feeds 24V DC bus via safety interlock relay block.
Battery and solar combiner include inline fusing and emergency stop loop.
Motor drivers [azimuth & elevation] share the 24V bus downstream of interlocks.

Control: Main controller distributes I2C and UART backbone to peripherals.
GPS receiver connects on dedicated UART; LCD status panel resides on I2C.
Joystick axes feed MCU analog inputs; pushbuttons wired as digital inputs.
Limit switches [AZ+/-, EL+/-] return to MCU GPIO with shielded cabling.
Motor encoders route through drivers for closed-loop feedback and MCU telemetry.

Safety: Interlock relay removes power from drivers when any E-stop is triggered.
Door switches and overcurrent sensors wired in series within interlock chain.
Ground fault detection bonded to chassis and controller shield reference.